

Operating Systems

Liu Fagui

Who am I?

- Prof. Fagui Liu (刘发贵)
- BUAA/SCUT
- Research: Operating Systems & Embedded Software, RFID & IoT, Intelligent Semantic & Service .
- Phone: 020-39380283-3402
- E-mail: fgliu@scut.edu.cn

Overview

- **Teaching Assistants**

- Zhou Xiaochang: 1006197308@qq.com , 13570210615
- Ou Jiamin: 414802725@qq.com , 13570208566

- **Textbook**

- 《现代操作系统》(中文版第三版), 机械工业出版社
- 《Modern Operating System》(英文版第三版), 机械工业出版社

- **参考文献**

- 《现代操作系统》(中文版第二版), 机械工业出版社
- 《计算机操作系统教程》(修订版), 西安电子科技大学出版社
- Tanenbaum AS, Operating System Design and Implementation, Prentice-Hall, 2003.

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Course Outline

- **Time-line of lectures and assignments**

Time-Due Topic	Material	Assignments
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1: Introduction, Overview	Ch.1 Reading	Homework 1
2: Processes/Threads	Ch.2.1,2.2,	
3: CPU scheduling	Ch.2.4	
4: Process Synchronization	Ch.2.3, 2.5	Homework 2
5: Deadlock	Ch.6	Lab 1& Test1 Homework 3
6: Memory Management	Ch.3.1,2,3,	
6~8: Virtual Memory	Ch.3.4~8	Lab2 & Test2 Homework 4
9~11:File Systems	Ch.4	Lab3 Homework 5
14~15:I/O Systems	Ch.5	Lab4 & Test3 Homework 6
16: Security	Ch.9	
17~18:Case Study	Ch. 10	
19:EXAM		
12-18: OS Design		

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Course Information

- Prerequisites
 - Hardware:
 - Software: Data structure, C++
- Laboratory Facilities
 - Linux

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Why are you here?

- Fulfill the requirement
- Prerequisite for other courses
 - Network
 - Distributed systems
 - Real-time systems
 - Multimedia systems
- Future plans
 - Graduate schools for CS
 - Employment

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About this course...

Principles

- System concepts
- OS design
- Some theory(理论)
- Rationale(基本原理)
- Practice

Goals

- Understand OS decisions
- Basis for future learning
- Control computer

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Expect (Some) Pain

Fast pace

Many hard material

Lots of Labs (programming)

But....



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Lecture Format

- Help you understand important and hard OS concepts
 - Group discussions
 - Animation (动画演示)
 - Real life examples
- Lectures do not cover everything
 - Not all questions in quizzes or exams are from lectures
- Students responsibility
 - Attend lectures
 - Read textbooks
 - Quiz, Lab, Homework, Exam

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Why so many grades?

Intended to reduce pressure/anxiety

More data points = less noise

- Final exam: 60%
- Discussions & Quizzes: 4%
- Homework & Test: 20%
- Labs: 16%

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Chapter 1

Introduction

Chapter Outline

- What is OS? What does it do?
- History of OS
- The Operating System Zoo
- Computer Hardware overview
- Operating System Concepts
- System Calls
- Operating Systems Structure
- Summary

What Is an OS? Real life example?

- 2-3 persons group discussion
- 2 minutes

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What Is An Operating System (1)

A modern computer consists of:

- One or more processors,
- Main memory
- Disks
- Printers
- Various input/output devices

Managing all these components requires a layer of software – the **operating system**

What Is An Operating System (2)

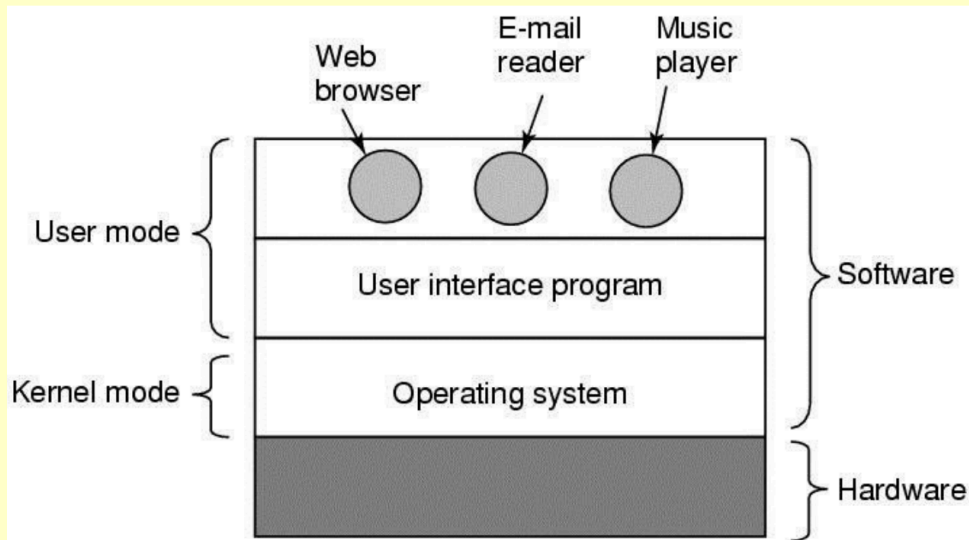


Figure 1-1. Where the operating system fits in.

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What Is an OS?

Resources

- Allocation(分配)
- Protection(保护)
- Reclamation(回收)
- Virtualization(虚拟化)

Services

- Abstraction(提取)
- Simplification(简化)
- Convenience(方便)
- Standardization(标准化)

Makes computers simpler

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What Is an OS?

Resources

- Allocation
- Protection
- Reclamation
- Virtualization

Finite resources

Competing demands

Examples:

- CPU
- Memory
- Disk
- Network

Government

Limited budget,
Land,
Oil,
Gas,

Linux or Windows?

Democrat or Republic(民主党党员或共和国)?

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What Is an OS?

Resources

- Allocation
- **Protection**
- Reclamation
- Virtualization

You can't hurt me

I can't hurt you

Implies some degree of
safety & security

Government

Law and order

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What Is an OS?

Resources

- Allocation
- Protection
- **Reclamation**
- Virtualization

The OS giveth

The OS taketh away

Voluntary(自愿的) at run time

Implied at termination

Involuntary(自然而然的)

Cooperative

Government

Income Tax

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After this lecture...

- Reading assignment: chapter 1
- Start next week with Chapter 2
- Homework:
 - P44-45: 1,10,12, 15, 20

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Thank you!
Q&A

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